# Design-Stage Balance Control: Replicating and Extending CEM

Response to Iacus, King, and Porro (2012), *Causal Inference without Balance Checking: Coarsened Exact Matching*, doi:10.1093/pan/mpr013.


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## Abstract.

I replicate the paper's central method comparison using the LaLonde-based Monte Carlo design and extend it with coarsening sensitivity, perturbation and placebo diagnostics, and a comparison with entropy balancing (EBAL). In the reproduced output, CEM has lower L1 imbalance than Mahalanobis, propensity score, and genetic matching (0.382 vs. 0.539, 0.615, and 0.562), and weighted CEM has the lowest RMSE. The extensions sharpen rather than reverse that result: under perturbation, weighted CEM has the highest MSPE ratio, but it still has the lowest absolute MSPE in both baseline and perturbed settings. The findings support the paper's design-stage argument and suggest that applied researchers should report both ratio-based and level-based robustness metrics.



causal inference | matching | CEM | observational studies | treatment effects | entropy balancing | replication | balance diagnostics | robustness | Monte Carlo simulation

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## 1. Introduction and Response Objective

This paper evaluates how far the original case for CEM holds once the core replication is extended. I keep the article's design-stage emphasis but shift the contribution from restatement to stress-testing the original claims: the extensions examine coarsening sensitivity, perturbation and placebo diagnostics, and comparison with entropy balancing. The objective is therefore narrower than the original article's broad methodological defense. It asks not whether CEM can work, but how stable its apparent advantages remain when the design is stressed in ways applied researchers would reasonably care about.

## 2. Summary of the Original Paper

Iacus, King, and Porro, "Causal Inference without Balance Checking: Coarsened Exact Matching" (*Political Analysis*, 2012), study a causal question: how should researchers preprocess observational data so that treated and control units are comparable before estimating treatment effects? The paper argues that many matching procedures preserve sample size or optimize fit without guaranteeing the balance needed for credible causal inference. The phrase "without balance checking" captures the authors' central criticism of iterative workflows that search across specifications without an ex ante standard for acceptable imbalance.

Their proposed solution is coarsened exact matching within the monotonic imbalance bounding class.[1] Analysts choose acceptable imbalance ex ante through substantively meaningful coarsening choices rather than tuning the design after inspecting results. The paper's contribution is therefore both methodological and procedural: it offers an algorithm, but it also offers a workflow in which design decisions are made before outcome analysis and in which the trade-off between balance and retention is explicit enough for readers to evaluate. Two passages make that argument explicit. First, the authors write, "You get what you want rather than getting what you get." Second, they argue that in observational studies "bias is more crucial than variance." Together, those statements summarize the paper's defense of ex ante balance control.

## 3. Data, Code, and Access

All data and code used in this project are publicly available through the Harvard Dataverse replication archive associated with the article. The materials used here draw on the LaLonde-based simulation objects and scripts in that archive, including the `cemsim-penta` workflow and related measurement-error components.

## 4. Replication Focus and Methods

The main replication centers on the paper's method comparison and associated balance ranking. The methods compared are Mahalanobis matching (MAH), propensity score matching (PSC), genetic matching (GEN), and CEM in both unweighted and weighted variants. The evaluation metrics are bias, standard deviation, RMSE, runtime, and L1 imbalance. For readers outside the matching literature, the practical interpretation is simple: L1 tells us how similar treated and control covariate distributions are after matching,[2] while RMSE summarizes estimator quality by combining bias and dispersion.

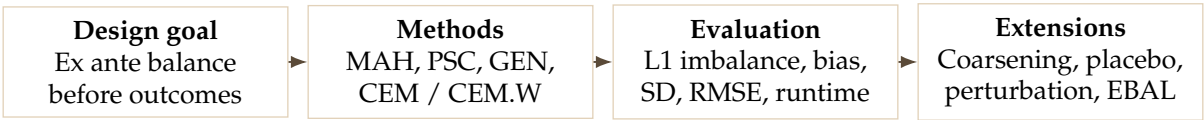



**Figure 1:** *Study roadmap.*

[1] Coarsened exact matching groups observations into substantively meaningful bins, such as age ranges or income ranges, and then matches treated and control units within those bins.

[2] The L1 measure is a summary of how different the treated and control groups look across the joint distribution of observed covariates. Lower values indicate better balance.

## 5. Replication Results

The replication succeeds and supports the paper's core empirical claim. In the reproduced output, CEM has much lower L1 imbalance than MAH, PSC, and GEN (0.382 vs. 0.539, 0.615, and 0.562). Weighted CEM also has the lowest RMSE in this run (111.4 vs. 1077.2, 1058.3, and 508.3 for MAH, PSC, and GEN).

The result holds across metrics. The frontier plot, the direct L1 comparison, and the bias–SD profile all point in the same direction: CEM's design-stage restriction yields a large improvement in covariate balance without producing the variance inflation or instability that would weaken the causal-design argument.

**Table 1:** *Core replicated simulation table with bias, SD, RMSE, matched counts, runtime, and L1 by method.*

| Method | Bias | SD | RMSE | Treated | Controls | Seconds | L1 |
|---|---|---|---|---|---|---|---|
| RAW | -423.72 | 1566.49 | 1622.63 | 151 | 293 | 0.00 | 0.640 |
| MAH | 784.80 | 737.93 | 1077.20 | 151 | 151 | 0.02 | 0.539 |
| PSC | 260.45 | 1025.83 | 1058.28 | 151 | 151 | 0.02 | 0.615 |
| GEN | 78.93 | 502.22 | 508.33 | 151 | 143 | 18.69 | 0.562 |
| CEM | 153.53 | 136.58 | 205.48 | 86 | 151 | 0.23 | 0.382 |
| CEM.W | 0.78 | 111.39 | 111.38 | 86 | 151 | 0.23 | 0.382 |

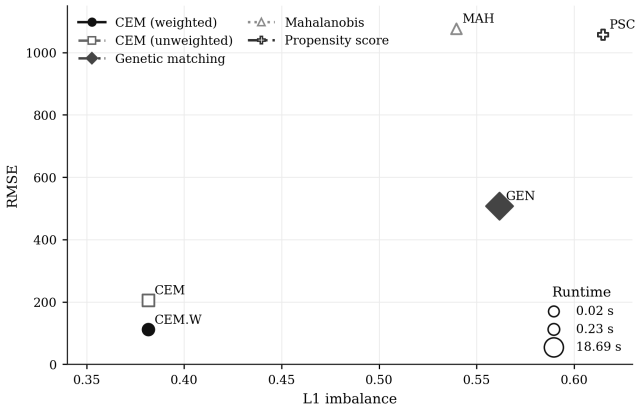
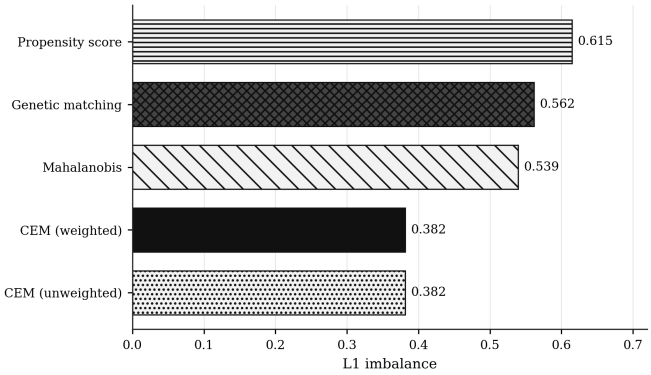
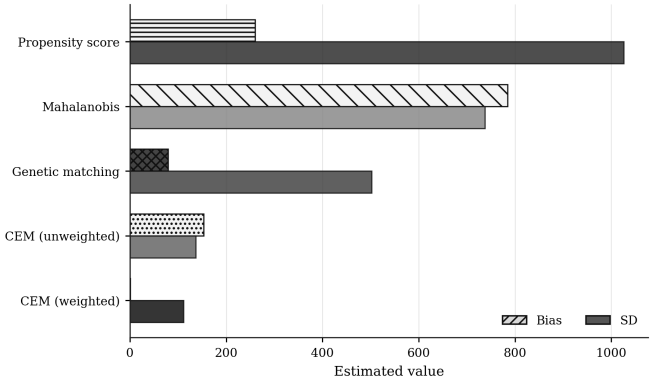



**Figure 2:** *Joint frontier for balance, accuracy, and runtime. Methods farther left are better balanced, methods lower on the page have lower RMSE, and larger points take longer to run.*

Weighted CEM achieves these gains while remaining computationally light, and the smaller matched sample under CEM reflects an explicit bias–retention trade-off rather than an opaque tuning exercise. In this replication, that trade-off is favorable: CEM.W is the most balanced method and also the most accurate by RMSE, while unweighted CEM remains markedly better than the non-CEM alternatives.




**(a)** *Post-match L1 imbalance across the core methods. Lower values indicate better covariate balance, and CEM performs best in this comparison.*




**(b)** *Bias and standard deviation across the same methods. CEM substantially reduces both quantities relative to the unmatched data and compares favorably with the alternatives.*

**Figure 3:** *Additional core replication visuals. These panels separate the balance comparison from the bias–dispersion comparison while reinforcing the same ranking pattern seen in the table and frontier plot.*

## 6. Extension 1: Coarsening-Sensitivity Frontier

This extension asks how much CEM results change when the coarsening rule used before matching is adjusted. I estimated four specifications, moving from finer to coarser binning, and plotted each one on a coarsening-sensitivity frontier that links post-match balance to retained sample size. The pattern is clear. As coarsening becomes less strict, the matched sample increases from 398 to 608 units,[3] while post-match L1 remains within a fairly tight range of about 0.223 to 0.248. The exercise makes the balance–retention trade-off explicit.[4]

**Table 2:** *Specification-level sensitivity table for CEM coarsening choices and resulting balance-retention outcomes.*

| Spec | Age | Edu | Re74 | Re75 | Matched N | L1 matched |
|---|---|---|---|---|---|---|
| S1 | 6 | 5 | 6 | 6 | 398 | 0.248 |
| S2 | 5 | 4 | 5 | 5 | 468 | 0.238 |
| S3 | 4 | 4 | 4 | 4 | 498 | 0.246 |
| S4 | 3 | 3 | 3 | 3 | 608 | 0.223 |

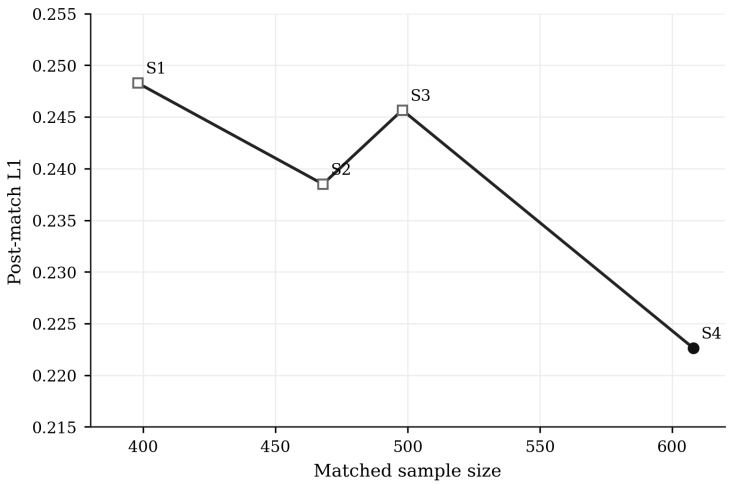



**Figure 4:** *Coarsening-sensitivity frontier mapping post-match L1 imbalance against retained matched sample size. Each point corresponds to one specification in Table 2; moving right along the curve trades finer coarsening for a larger matched sample.*

Operationally, the four specifications were implemented by progressively reducing the number of quantile-based bins for age, education, re74, and re75 from $(6,5,6,6)$ to $(3,3,3,3)$, while holding the matching framework and estimand fixed. This yields a controlled sensitivity design in which only coarsening granularity is allowed to vary.[5] For each specification, the same coarsening structure used to construct matches was also used to evaluate post-match imbalance, and matched treated/control totals were recorded. The comparison therefore isolates the design effect of coarsening from changes in model form or target parameter.

The estimated frontier indicates that, over the tested range, coarsening mainly shifts retention rather than generating large balance deterioration. The increase in matched sample size from 398 to 608 is substantial relative to the narrower movement in post-match L1 (0.223 to 0.248), implying a comparatively flat local trade-off in this region.[6] In substantive terms, the extension supports the stability of the CEM design under moderate coarsening changes and motivates treating coarsening as a calibration parameter aligned with overlap and precision objectives.

[3] Matched sample size is the number of observations that remain after units without acceptable matches are removed.

[4] A frontier is not a single "correct" answer. It is a map of feasible design choices, showing what is gained and what is given up at each setting.

[5] Quantile-based coarsening defines bins from empirical percentiles, so each bin contains a comparable share of observations, subject to ties in the data.

[6] A flatter frontier indicates that meaningful gains in retained sample size can be achieved with limited movement in the imbalance metric over the tested specifications.

## 7. Extension 2: MSPE-Ratio Diagnostics and Placebo Reweighting

I implement perturbation-based MSPE diagnostics and placebo reassignment checks to evaluate ranking stability under stress.[7] The diagnostics separate proportional deterioration from absolute error magnitude. Under perturbation, CEM.W records the largest relative increase in MSPE (ratio = 2.191), yet it retains the lowest absolute MSPE in both baseline and perturbed settings (12,084.65 and 26,476.79, respectively).

**Table 3:** *Baseline and perturbed MSPE by method, with perturbed-to-baseline ratios. Ratio movements are interpreted jointly with MSPE levels.*

| Method | MSPE base | MSPE perturbed | Ratio |
|---|---|---|---|
| CEM.W | 12,084.65 | 26,476.79 | 2.191 |
| PSC | 967,081.52 | 1,146,387.41 | 1.185 |
| MAH | 1,080,647.65 | 1,215,051.41 | 1.124 |
| EBAL | 204,145.73 | 252,352.65 | 1.236 |

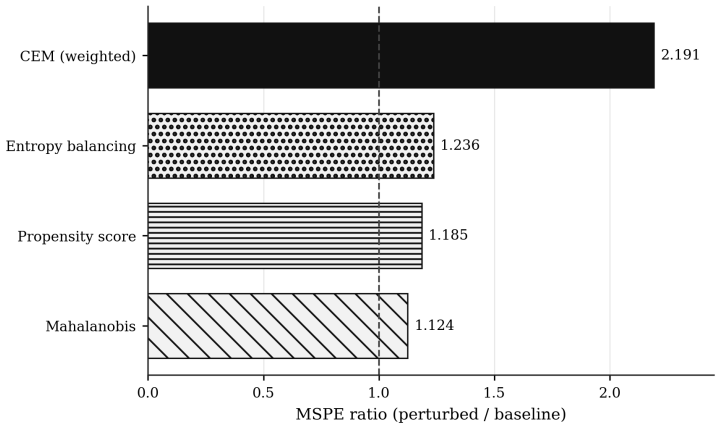



**Figure 5:** *Perturbed-to-baseline MSPE ratios by method. Values above one indicate deterioration under perturbation.*

**Table 4:** *Placebo reassignment summary reporting observed ATT, placebo mean, placebo SD, and two-sided placebo* p*-values.*

| Method | Observed ATT | Placebo mean | Placebo SD | $p$ |
|---|---|---|---|---|
| CEM.W | 550.96 | -50.31 | 692.73 | 0.428 |
| PSC | 1022.61 | 45.28 | 516.77 | 0.044 |
| MAH | 677.54 | 119.97 | 513.92 | 0.204 |
| EBAL | 872.87 | -9.37 | 494.38 | 0.072 |

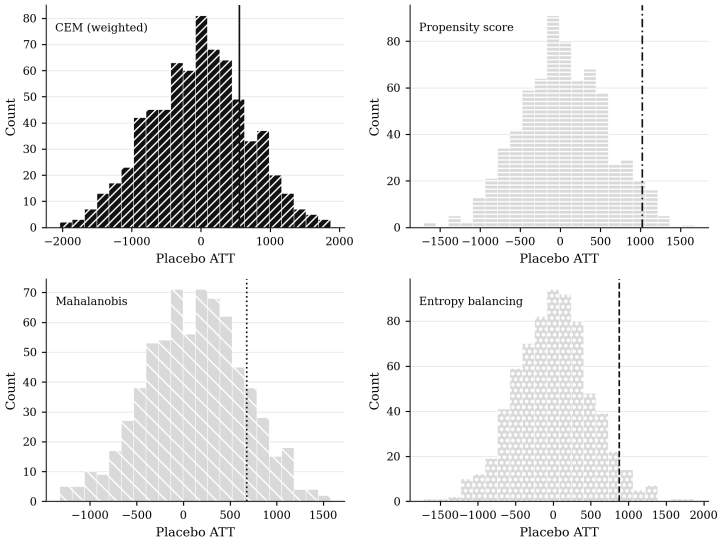



**Figure 6:** *Method-specific distributions of placebo ATT estimates from randomized treatment reassignment, with each observed ATT shown as a vertical reference line. A method whose observed ATT sits well outside the placebo distribution is less likely to be confounded by chance assignment.*

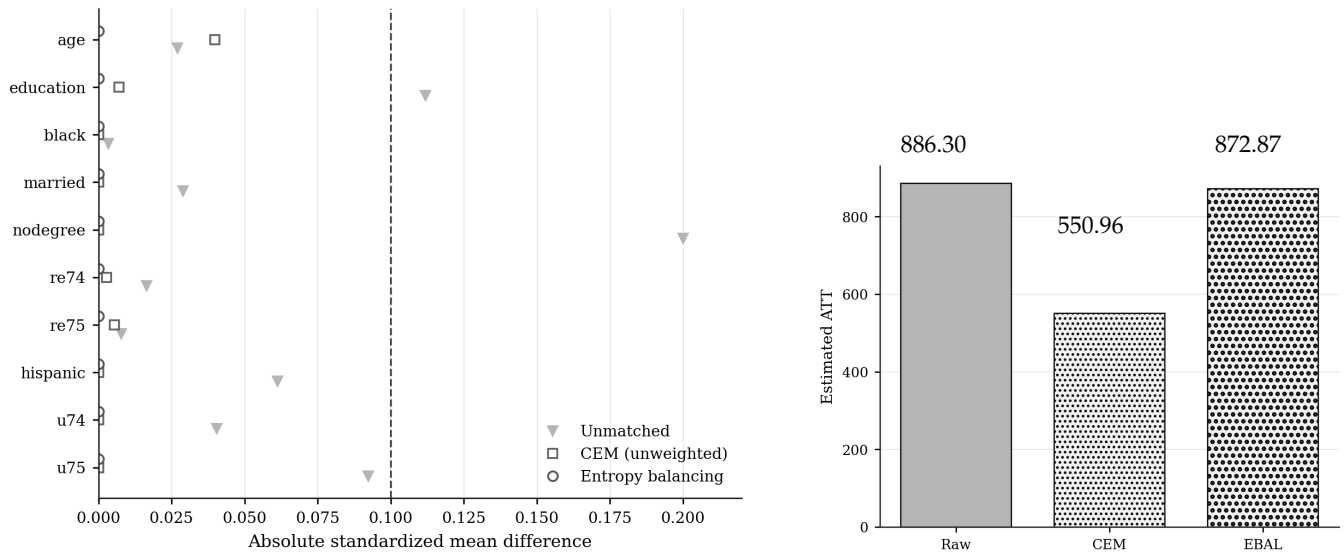
Extension 2 shows that robustness rankings depend on whether evaluation is proportional or absolute. CEM.W appears most sensitive in ratio terms because its baseline MSPE is very low, while competing methods exhibit smaller ratios but substantially larger error levels in both baseline and perturbed states. The placebo distributions reinforce heterogeneity in null behavior across estimators, with PSC the only method that rejects the null under placebo reassignment in the tabulated $p$-values. Robustness interpretation is therefore best anchored in a joint profile of MSPE levels, deterioration ratios, and placebo evidence rather than any single diagnostic.

[7] A placebo reassignment test randomly reassigns treatment labels and evaluates whether a method continues to produce large estimated effects when the assignment is artificial.

## 8. Extension 3: CEM Versus Entropy Balancing

I compare CEM and entropy balancing (EBAL) under the same observational setup, covariate set, and ATT target to evaluate design sensitivity across balancing strategies.[8] This comparison is informative because CEM and EBAL enforce balance through different mechanisms: CEM through coarsened strata and matched support, EBAL through continuous reweighting.

Both approaches substantially improve observed balance relative to the raw sample. Mean absolute standardized mean differences decline from 0.0590 in the raw data to 0.0055 under CEM and to values that are effectively zero under EBAL after rounding (covariate-level EBAL magnitudes are on the order of $10^{-6}$). However, ATT estimates remain materially different across designs (Raw 886.3, CEM 551.0, EBAL 872.9), indicating that balance improvement does not mechanically imply convergence in effect size estimates when assignment is nonrandom.




**(a)** *Love plot of* absolute *standardized mean differences (|SMD|) for Raw, CEM, and EBAL. The panel reports imbalance magnitudes only (non-negative values), with no sign direction; smaller values indicate better balance.*

**(b)** *ATT point estimates for Raw, CEM, and EBAL. The panel links balancing design to substantive effect magnitude.*

**Figure 7:** *Extension 3 comparison of CEM and entropy balancing. The left panel reports covariate imbalance magnitudes (|SMD|), and the right panel reports corresponding ATT estimates. Covariate-level balance values are reported in the appendix table.*

The extension indicates that strong observed balance is necessary but not sufficient for cross-design agreement in treatment effects. CEM and EBAL reduce imbalance through different weighting and support rules, which generate different effective comparison populations. The gap between their ATT estimates (551.0 vs. 872.9) is best interpreted as design-induced estimand sensitivity under limited overlap: CEM restricts the analysis to regions of common support and thereby shifts the effective estimand, while EBAL reweights without trimming and remains closer to the full-sample average. That distinction is a feature of the designs, not a diagnostic failure.

Additional diagnostic visuals are collected in Appendix A so the main text can stay centered on the core replication and extension comparisons. In particular, the error-distribution ECDF and cross-metric rank profile now appear in Figure 8 alongside the broader supplementary figure set.

[8] Entropy balancing reweights control units so that selected covariate moments align with those of treated units.

## 9. Integrated Discussion

The reproduced comparison strongly favors CEM on the original benchmark: it delivers the lowest post-match L1 among the core methods and, in weighted form, the lowest RMSE. The extensions confirm that this advantage is not specific to a single specification; it holds under coarsening variation, perturbation, and an expanded method comparison.

Extension 1 demonstrates that moderate changes in coarsening granularity mainly reallocate matched sample retention rather than sharply degrading balance. Over the tested range, the frontier is relatively flat: matched sample size rises substantially while post-match L1 moves only modestly. This supports the view that coarsening is best understood as a calibration parameter within the design problem rather than as an arbitrary tuning choice detached from substantive overlap and precision considerations.

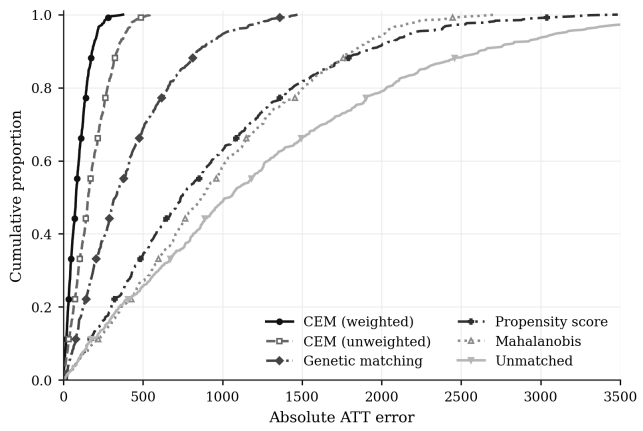
Extension 2 adds a more demanding stress test. Under perturbation, weighted CEM deteriorates the most in proportional terms, yet it remains the best method in absolute MSPE. The placebo reassignment results point in the same direction: estimator behavior under stress is heterogeneous, and no single diagnostic is sufficient on its own. A method can look fragile on a proportional metric precisely because it starts from a very low baseline error level.

Extension 3 places CEM alongside entropy balancing on the same observational setup. Both methods substantially reduce covariate imbalance, but ATT estimates diverge materially across Raw (886.3), CEM (551.0), and EBAL (872.9). This divergence is a design-level feature rather than a diagnostic anomaly: different balancing procedures define different effective comparison populations, and the gap between CEM and EBAL reflects that difference in estimand rather than a failure of either method's balance procedure.

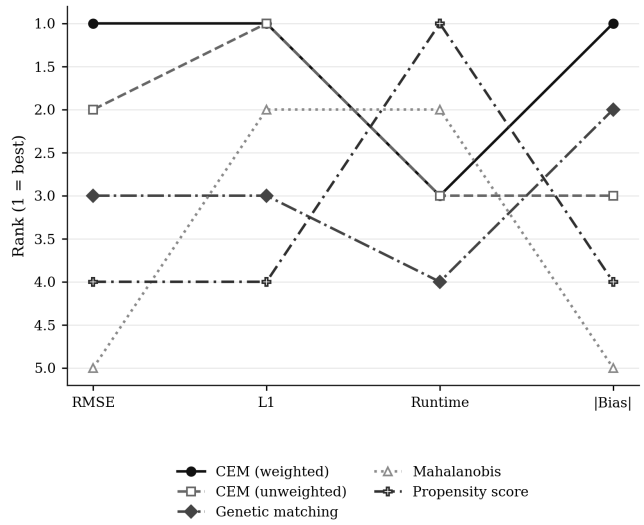
Across all three extensions, the original article's emphasis on ex ante balance control remains persuasive, but the most defensible statement of its claims is procedural rather than absolute. CEM performs very well in the replicated benchmark, retains that strength under moderate coarsening variation, and remains highly competitive under perturbation and placebo diagnostics. At the same time, the extensions show why applied evaluation should not stop at a single balance measure or a single matched specification. A convincing design-based analysis should report how conclusions change across admissible coarsenings, how estimators behave under stress, and how alternative balancing frameworks alter both balance and effect estimates.

The conclusion is supportive but more disciplined than a simple restatement of the original article. Design-stage balance control is a first-order requirement for credible causal inference in observational data, and CEM remains a powerful implementation of that principle. The added analyses show where that conclusion is stable, where it depends on the metric used, and why no single diagnostic (whether a ratio, an error level, or a placebo $p$-value) is sufficient on its own. The extensions do not overturn the original result; they map its boundaries and clarify the conditions under which it holds.

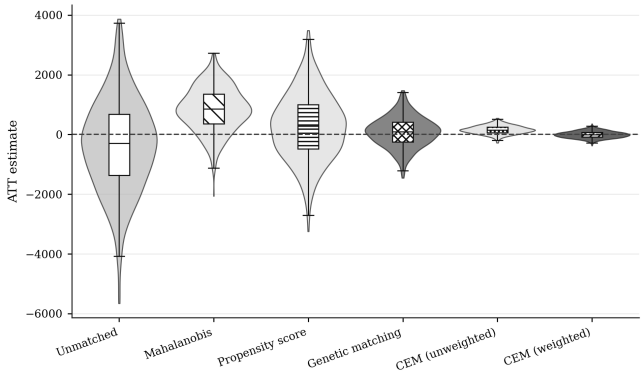
# Appendix A: Supplemental Figures



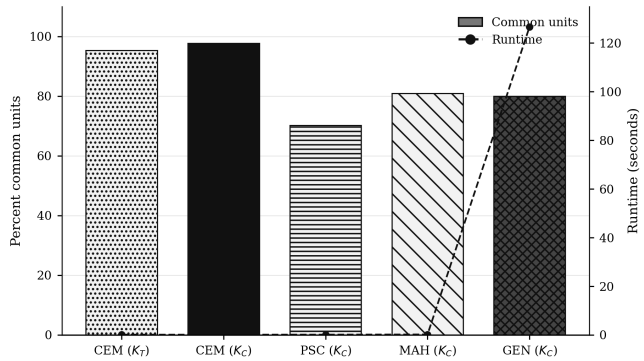

**(a)** *Empirical CDF of absolute ATT error across methods, showing the full distribution of estimation error rather than only summary moments.*



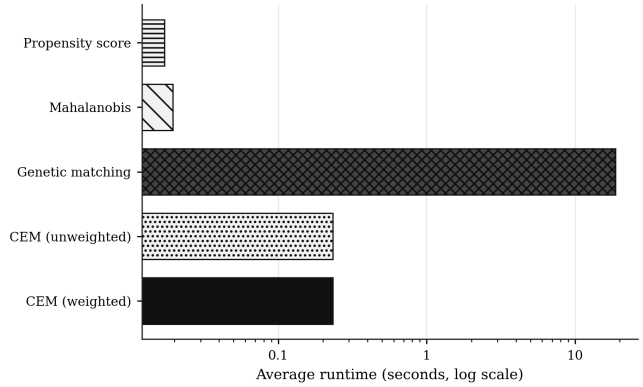

**(b)** *Cross-metric rank profile comparing performance across balance, accuracy, and computational criteria.*



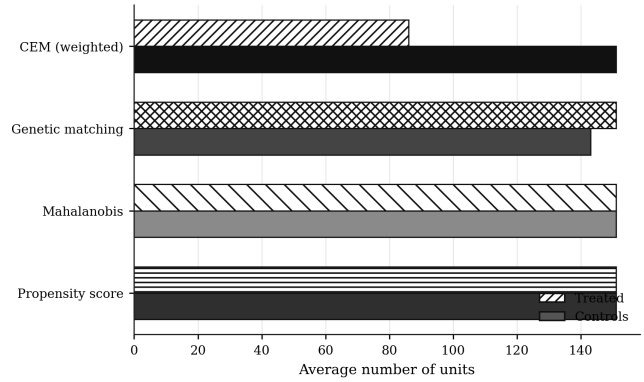

**(c)** *Violin-and-box distribution of ATT estimates by method with the true ATT as a reference line.*




**(d)** *Measurement-error stability diagnostic combining common-unit retention with runtime across methods.*

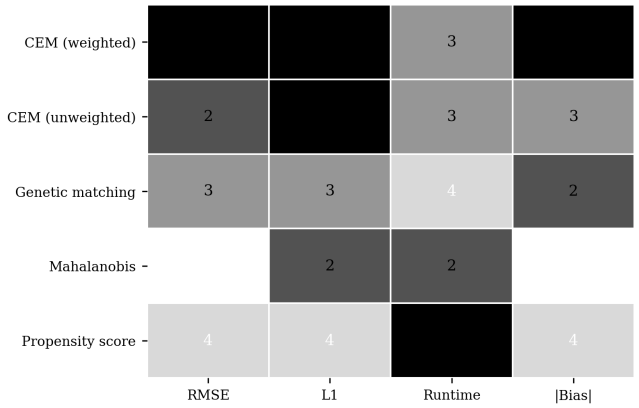



**(e)** *Runtime comparison on a log scale, emphasizing the large computational differences across preprocessing strategies.*

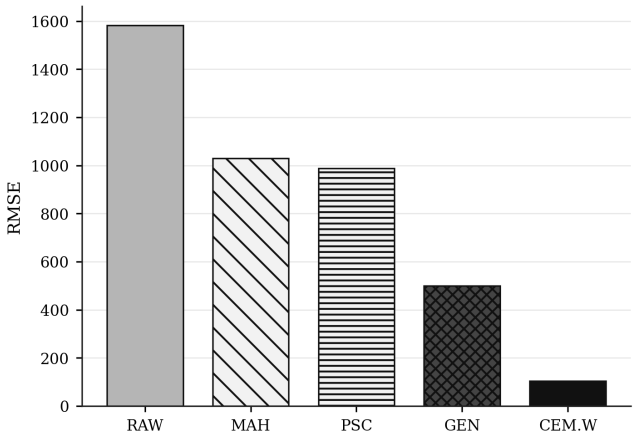



**(f)** *Average treated and control counts retained by method, clarifying the balance–retention trade-off.*

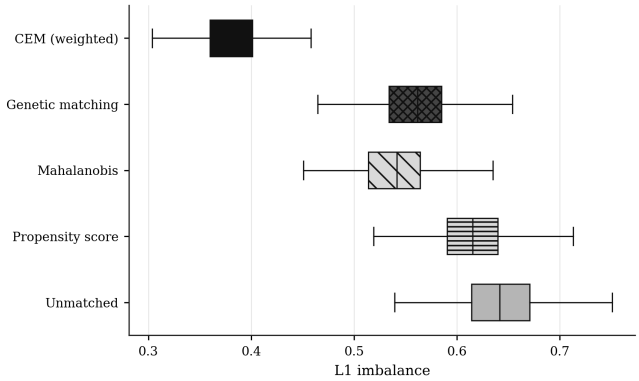
**Figure 8:** *Appendix A overview diagnostics. The three paired rows summarize overall error behavior, estimator distributions, measurement-error stability, and computational or sample-composition trade-offs for the replication designs.*



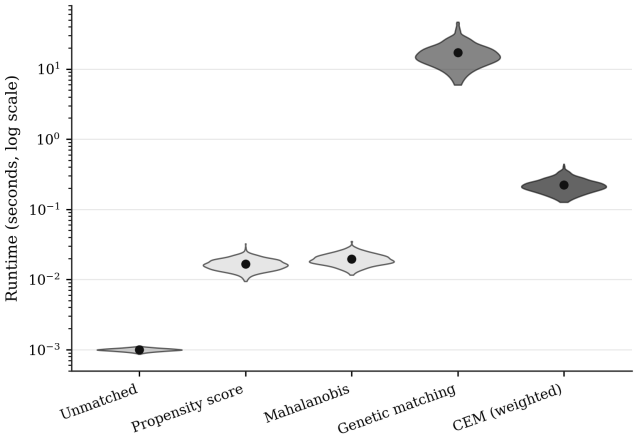

**(g)** *Heatmap of metric-by-method ranks, offering a compact summary of relative performance across criteria.*




**(h)** *Reduced re-simulation RMSE comparison used to verify that the main ordering is not an artifact of a single run.*




**(i)** *Distribution of L1 imbalance across simulation draws, showing variation around the central balance ranking.*




**(j)** *Runtime distribution across simulation draws, highlighting both central tendency and spread in computational burden.*

**Figure 8:** *Appendix A supplementary robustness panels. These paired figures extend the main text with rank aggregation, reduced re-simulation validation, and distributional diagnostics for both balance and runtime.*

## Appendix B: Supplemental Tables

**Table 5:** *Matching method performance under measurement error: share of common units retained and wall-clock runtime (seconds).*

| Metric | CEM($K_T$) | CEM($K_C$) | PSC($K_C$) | MAH($K_C$) | GEN($K_C$) |
|---|---|---|---|---|---|
| Common units (%) | 95.3 | 97.7 | 70.2 | 80.9 | 80.0 |
| Runtime (s) | 0.07 | 0.07 | 0.08 | 0.15 | 126.64 |

**Table 6:** *Full simulation results: bias, standard deviation, RMSE, wall-clock runtime, and L1 imbalance across all methods. CEM (weighted) achieves near-zero bias at a fraction of the cost of genetic matching.*

| Method | Bias | SD | RMSE | Runtime (s) | L1 | Label |
|---|---|---|---|---|---|---|
| RAW | -423.72 | 1566.49 | 1622.63 | 0.000 | 0.640 | Unmatched |
| MAH | 784.80 | 737.93 | 1077.20 | 0.020 | 0.539 | Mahalanobis |
| PSC | 260.45 | 1025.83 | 1058.28 | 0.017 | 0.615 | Propensity score |
| GEN | 78.93 | 502.22 | 508.33 | 18.688 | 0.562 | Genetic matching |
| CEM | 153.53 | 136.58 | 205.48 | 0.234 | 0.382 | CEM (unweighted) |
| CEM.W | 0.78 | 111.39 | 111.38 | 0.234 | 0.382 | CEM (weighted) |

**Table 7:** *Standardized mean differences (SMD) by covariate for the raw sample, CEM, and entropy balancing (EBAL). Values near zero indicate better covariate balance.*

| Covariate | Raw | CEM | EBAL |
|---|---|---|---|
| age | 0.027 | -0.040 | -0.000 |
| education | 0.112 | -0.007 | 0.000 |
| black | 0.003 | 0.000 | -0.000 |
| married | 0.029 | 0.000 | 0.000 |
| nodegree | -0.200 | 0.000 | 0.000 |
| re74 | -0.017 | -0.003 | -0.000 |
| re75 | 0.008 | -0.005 | -0.000 |
| hispanic | -0.061 | 0.000 | -0.000 |
| u74 | -0.040 | 0.000 | -0.000 |
| u75 | -0.092 | 0.000 | -0.000 |

**Table 8:** *Monte Carlo validation (1,000 simulations): bias, standard deviation, and RMSE for each method. CEM (weighted) consistently achieves the lowest error across all draws.*

| Method | Bias | SD | RMSE |
|---|---|---|---|
| RAW | -451.56 | 1523.08 | 1582.51 |
| MAH | 804.41 | 646.19 | 1030.13 |
| PSC | 309.18 | 942.89 | 988.55 |
| GEN | 92.50 | 493.92 | 500.48 |
| CEM.W | 0.52 | 103.92 | 103.49 |

## Appendix C: AI Statement

I used OpenAI Codex to generate and refine most parts of the replication workflow, including scripts used to organize outputs and produce the plots reported in the paper and appendix. I also used Anthropic's sonnet model in a limited editorial role to help refine wording in the abstract and the integrated discussion section. All claims, design choices, numerical checks, and final interpretations were validated by me, and I take responsibility for the final text, analysis, and presentation.

## References


[1] Abadie, A., & Imbens, G. W. (2006). Large sample properties of matching estimators for average treatment effects. *Econometrica, 74*(1), 235–267. https://doi.org/10.1111/j.1468-0262.2006.00655.x

[2] Diamond, A., & Sekhon, J. S. (2013). Genetic matching for estimating causal effects: A general multivariate matching method for achieving balance in observational studies. *Review of Economics and Statistics, 95*(3), 932–945. doi:10.1162/REST_a_00318

[3] Hainmueller, J. (2012). Entropy balancing for causal effects: A multivariate reweighting method to produce balanced samples in observational studies. *Political Analysis, 20*(1), 25–46. https://doi.org/10.1093/pan/mpr025

[4] Ho, D. E., Imai, K., King, G., & Stuart, E. A. (2007). Matching as nonparametric preprocessing for reducing model dependence in parametric causal inference. *Political Analysis, 15*(3), 199–236. https://doi.org/10.1093/pan/mpl013

[5] Iacus, S. M., King, G., & Porro, G. (2011a). Multivariate matching methods that are monotonic imbalance bounding. *Journal of the American Statistical Association, 106*(493), 345–361. https://doi.org/10.1198/jasa.2011.tm09599

[6] Iacus, S. M., King, G., & Porro, G. (2011b). Replication data and code for CEM article [Data set]. Harvard Dataverse. doi:10.7910/DVN/NMMYYW

[7] Iacus, S. M., King, G., & Porro, G. (2012). Causal inference without balance checking: Coarsened exact matching. *Political Analysis, 20*(1), 1–24. https://doi.org/10.1093/pan/mpr013

[8] King, G., & Nielsen, R. (2019). Why propensity scores should not be used for matching. *Political Analysis, 27*(4), 435–454. https://doi.org/10.1017/pan.2019.11

[9] LaLonde, R. J. (1986). Evaluating the econometric evaluations of training programs with experimental data. *American Economic Review, 76*(4), 604–620.

[10] Rosenbaum, P. R., & Rubin, D. B. (1983). The central role of the propensity score in observational studies for causal effects. *Biometrika, 70*(1), 41–55. https://doi.org/10.1093/biomet/70.1.41

[11] Stuart, E. A. (2010). Matching methods for causal inference: A review and a look forward. *Statistical Science, 25*(1), 1–21. https://doi.org/10.1214/09-STS313